🎯 MISMATHE

Physics Practice Bank — v = u + at in Motion in a Plane

Every CET question type on this formula · Easy → Medium → Hard · Class 11

📘 Chapter: Motion in a Plane | 🎯 Prep for RT-03 · Sun 16 Aug 2026 | 🧮 Take g = 10 m/s² unless stated | 📝 38 Questions · 3 Difficulty Tiers + Full Answer Key

CONCEPT RECAP

$\vec{v} = \vec{u} + \vec{a}t$ $v_x = u_x + a_xt$ $v_y = u_y + a_yt$

- Valid **only** when acceleration $\vec{a}$ is constant in **both magnitude and direction** over the interval.
- In a plane, treat x and y as two **independent 1-D motions** — solve each axis separately, then combine as vectors.
- Commonest CET use: **projectile motion** → $a_x = 0$, $a_y = -g$ (taking up as +y). Horizontal velocity never changes.
- On a v–t graph, $\vec{a}$ is the **slope**; on an a–t graph, the **area** under the curve gives the change in velocity — same equation, two readings.
- ⚠️ NOT directly valid over a finite interval for **uniform circular motion** — |a| is constant but its direction keeps rotating.

QUESTION TYPES COVERED IN THIS BANK

Direct substitution · Solve for u / a / t · Multi-stage (piecewise) motion · Simultaneous equations · v–t & a–t graph reading · Horizontal projectile · Angled projectile (components) · Time of flight / max height · Complementary-angle relations · Symmetric velocity relations · Circular motion validity · 2-D vector algebra · Relative velocity + acceleration · Assertion–Reason · Find-the-incorrect-statement

WORKED EXAMPLE — METHOD TO FOLLOW

Q. A particle is projected with u = 10 m/s at 30° above the horizontal. Find its velocity vector at t = 1 s. (g = 10 m/s², sin30° = 0.5, cos30° = 0.866)

1. Split into components: $u_x = 10 \times 0.866 = 8.66$ m/s, $u_y = 10 \times 0.5 = 5$ m/s
2. Apply $v_x = u_x + a_xt$ → $a_x = 0$, so $v_x = 8.66$ m/s (unchanged)
3. Apply $v_y = u_y + a_yt$ → $v_y = 5 + (-10)(1) = -5$ m/s
4. Recombine: $\vec{v} = 8.66\hat{i} - 5\hat{j}$ m/s → speed = $\sqrt{(8.66^2 + 5^2)} = \sqrt{100}$ = **10 m/s**, directed 30° below horizontal

Tier 1 · EASY

Q E1–E10 · straight substitution & graph basics

E1. u = 5 m/s, a = 3 m/s², t = 4 s. Find v.
- **a)** 14 m/s
- **b)** 17 m/s
- **c)** 20 m/s
- **d)** 23 m/s

E2. u = 8 m/s, a = 2 m/s², v = 20 m/s. Find t.
- **a)** 4 s
- **b)** 5 s
- **c)** 6 s
- **d)** 7 s

E3. v = 30 m/s, a = 5 m/s², t = 4 s. Find u.
- **a)** 5 m/s
- **b)** 10 m/s
- **c)** 15 m/s
- **d)** 20 m/s

E4. u = 10 m/s, v = 40 m/s, t = 6 s. Find a.
- **a)** 3 m/s²
- **b)** 5 m/s²
- **c)** 7 m/s²
- **d)** 9 m/s²

E5. A body decelerates: u = 25 m/s, a = −5 m/s². Its velocity at t = 3 s is:
- **a)** 5 m/s
- **b)** 10 m/s
- **c)** 15 m/s
- **d)** 20 m/s

E6. **$u = 18$ m/s, $a = -6$ m/s². Time to come to rest:**

a) 2 s
b) 3 s
c) 4 s
d) 5 s

E7. **In $v = u + at$, the term "at" must carry the same unit as:**

a) displacement
b) velocity
c) acceleration
d) time

E8. **For uniformly accelerated motion, the velocity–time graph is:**

a) a curve
b) a straight line, slope = a
c) a horizontal line
d) a straight line, slope = u

E9. **A v–t graph is a straight line through (0 s, 4 m/s) and (3 s, 16 m/s). Its acceleration is:**

a) 2 m/s²
b) 3 m/s²
c) 4 m/s²
d) 5 m/s²

E10. **For the same graph (E9), the velocity at $t = 5$ s (extrapolated) is:**

a) 20 m/s
b) 22 m/s
c) 24 m/s
d) 26 m/s

Tier 2 · MEDIUM

Q M1–M14 · piecewise motion, projectile components, circular-motion concept

M1. **A particle has $u = 10$ m/s, $a = 3$ m/s² for the first 2 s, then $a = -1$ m/s² for the next 3 s. Its velocity at $t = 5$ s is:**

a) 10 m/s
b) 13 m/s
c) 16 m/s
d) 19 m/s

M2. **A particle under constant acceleration has $v = 12$ m/s at $t = 2$ s and $v = 24$ m/s at $t = 5$ s. Its u and a are:**

a) 4 m/s, 4 m/s²
b) 2 m/s, 4 m/s²
c) 4 m/s, 2 m/s²
d) 6 m/s, 3 m/s²

M3. **Using "area under a–t graph = Δv": a is constant at 5 m/s² for 4 s, starting from $u = 6$ m/s. Final velocity is:**

a) 20 m/s
b) 24 m/s
c) 26 m/s
d) 30 m/s

M4. **A ball is projected horizontally at 15 m/s off a tower. Its vertical velocity component after 3 s is:**

a) 20 m/s
b) 25 m/s
c) 30 m/s
d) 35 m/s

M5. **For the same ball (M4), its resultant speed at $t = 3$ s is closest to:**

a) 30 m/s
b) 33.5 m/s
c) 36 m/s
d) 40 m/s

M6. **A projectile is launched at 40 m/s at 30° above horizontal ($\sin 30° = 0.5$, $\cos 30° = 0.866$). Its (u_x, u_y) are:**

a) 34.6, 20 m/s
b) 20, 34.6 m/s
c) 30, 25 m/s
d) 28, 30 m/s

M7. **For the same projectile (M6), v_y at $t = 1.5$ s is:**

a) 0 m/s
b) 5 m/s
c) 10 m/s
d) 15 m/s

M8. **For the same projectile (M6), the time to reach maximum height is:**

a) 1.5 s
b) 2 s
c) 2.5 s
d) 3 s

M9. **For the same projectile (M6), the magnitude of v_y at $t = 3$ s is:**

a) 5 m/s
b) 10 m/s
c) 15 m/s
d) 20 m/s

M10. **Particle A starts from rest with $a = 4$ m/s²; particle B moves at a constant 10 m/s on the same line. At what time do they have equal velocity?**

a) 2 s
b) 2.5 s
c) 3 s
d) 3.5 s

M11. **A block moves up an incline with $u = 20$ m/s and retardation 4 m/s² along the incline. Its velocity at $t = 6$ s is:**

a) 4 m/s up the incline
b) 4 m/s down the incline
c) 0
d) 44 m/s down the incline

M12. **A car speeds up from 10 m/s to 30 m/s in 4 s, then holds constant speed for the next 5 s. Its velocity at $t = 9$ s is:**

a) 25 m/s
b) 30 m/s
c) 35 m/s
d) 40 m/s

M13. **For uniform circular motion, is $|\Delta\vec{v}|$ over a quarter revolution simply $|\vec{a}|\times t$?**

a) Yes, always
b) No — $\vec{a}$'s direction keeps changing, so $\vec{a}t \neq \Delta\vec{v}$ in general
c) Yes, but only for a full revolution
d) No such relation exists

M14. **In non-uniform circular motion, the tangential component obeys $v_t = u_t + a_t t$ provided:**

a) a_t is constant
b) the radius is large
c) the speed is constant
d) it never applies in circular motion

Tier 3 · HARD

Q H1–H14 · CET previous-year style, vectors, relative velocity, assertion-reason

Multi-step and conceptual traps. These are the ones that separate 70% from 90%+.

H1. Two projectiles launch with the same speed u — one at θ, the other at (90° − θ) — landing at the same range R, with flight times t_1, t_2. The correct relation is:

- **a)** $t_1t_2 = 2R/g$
- **b)** $t_1t_2 = R/g$
- **c)** $t_1 + t_2 = R/g$
- **d)** $t_1 - t_2 = 2u/g$

H2. A ball is projected at 53° with u = 25 m/s (sin53° = 0.8). The time at which v_y drops to exactly half of u_y is:

- **a)** 0.5 s
- **b)** 1 s
- **c)** 1.5 s
- **d)** 2 s

H3. A particle's velocity changes from $(3\hat{i} + 4\hat{j})$ m/s to $(7\hat{i} + 4\hat{j})$ m/s in 2 s under constant acceleration. That acceleration is:

- **a)** $2\hat{i}$ m/s²
- **b)** $4\hat{i}$ m/s²
- **c)** $2\hat{i} + 2\hat{j}$ m/s²
- **d)** $1\hat{i}$ m/s²

H4. If the same acceleration (H3) continues from the same start, the velocity at t = 5 s is:

- **a)** $13\hat{i} + 4\hat{j}$
- **b)** $10\hat{i} + 4\hat{j}$
- **c)** $13\hat{i}$
- **d)** $8\hat{i} + 4\hat{j}$

H5. A body starts at $\vec{u} = 5\hat{i}$ m/s with constant $\vec{a} = -10\hat{j}$ m/s². The time at which its velocity makes 45° with the horizontal is:

- **a)** 0.25 s
- **b)** 0.5 s
- **c)** 0.75 s
- **d)** 1.0 s

H6. Particle P: u_P = 5 m/s, a_P = 2 m/s². Particle Q: u_Q = 15 m/s, a_Q = 0 (same line, same direction). At what time do they have equal velocity?

- **a)** 3 s
- **b)** 4 s
- **c)** 5 s
- **d)** 6 s

H7. For the same P and Q (H6), using $v_{rel} = u_{rel} + a_{rel}t$, the relative velocity of P w.r.t. Q at t = 3 s is:

- **a)** 4 m/s (Q faster)
- **b)** 4 m/s (P faster)
- **c)** 0
- **d)** 10 m/s (Q faster)

H8. Assertion (A): $\vec{v} = \vec{u} + \vec{a}t$ can be applied separately to the perpendicular components of velocity in projectile motion.
Reason (R): Acceleration due to gravity acts only along the vertical direction, so its component along the horizontal is zero.

- **a)** Both A and R are true, and R correctly explains A
- **b)** Both A and R are true, but R does NOT explain A
- **c)** A is true, R is false
- **d)** A is false, R is true

H9. Assertion (A): $\vec{v} = \vec{u} + \vec{a}t$ is valid for a particle moving in a circle at constant speed, over any time interval.
Reason (R): In uniform circular motion, the acceleration has constant magnitude equal to v^2/r.

- **a)** Both A and R are true, and R correctly explains A
- **b)** Both A and R are true, but R does NOT explain A
- **c)** A is true, R is false
- **d)** A is false, R is true

H10. Which statement about $\vec{v} = \vec{u} + \vec{a}t$ is INCORRECT?

- **a)** It is a vector equation and can be resolved into any two mutually perpendicular components
- **b)** It holds even when acceleration varies with time, as long as its final value is used
- **c)** For 1-D motion it reduces to the familiar scalar form v = u + at
- **d)** It implies velocity changes linearly with time under constant acceleration

H11. A ball is dropped (u = 0) from a height at the same instant another is thrown straight up with u = 20 m/s from the ground beside it; both feel g = 10 m/s² downward. Their relative velocity:

- **a)** increases with time
- **b)** decreases with time
- **c)** remains constant at 20 m/s
- **d)** becomes zero at t = 2 s

H12. Two projectiles launch at the same speed at 30° and 60°. The ratio of their times of flight ($t_{30°} : t_{60°}$) is:

- **a)** 1 : √3
- **b)** √3 : 1
- **c)** 1 : 1
- **d)** 1 : 3

H13. A projectile launched at 40° returns to the launch height at time T. Using $v_y(T) = u_y - gT$ with $T = 2u_y/g$, its vertical velocity component at landing is:

- **a)** 0
- **b)** $-u_y$ (same magnitude, reversed)
- **c)** $+u_y$ (unchanged)
- **d)** depends on horizontal speed too

H14. **$\vec{u} = (2\hat{i} + 3\hat{j})$ m/s. For 4 s, $\vec{a} = (1\hat{i} - 2\hat{j})$ m/s²; then for the next 2 s, $\vec{a}' = (0\hat{i} + 1\hat{j})$ m/s². The final velocity vector is:**

a) $6\hat{i} - 3\hat{j}$

b) $6\hat{i} + 3\hat{j}$

c) $8\hat{i} - 5\hat{j}$

d) $6\hat{i} - 5\hat{j}$

ANSWER KEY + ONE-LINE WORKING

Q		Working
E1	**b**	v = 5 + 3×4 = 17 m/s
E2	**c**	20 = 8 + 2t → t = 6 s
E3	**b**	u = v − at = 30 − 20 = 10 m/s
E4	**b**	a = (40 − 10)/6 = 5 m/s²
E5	**b**	v = 25 − 5×3 = 10 m/s
E6	**b**	0 = 18 − 6t → t = 3 s
E7	**b**	"at" must have units of velocity for the equation to be dimensionally consistent
E8	**b**	v = u + at is linear in t → straight line, slope = a
E9	**c**	a = (16 − 4)/3 = 4 m/s²
E10	**c**	v = 4 + 4×5 = 24 m/s
M1	**b**	v(2s) = 10+3×2 = 16; then 16 − 1×3 = 13 m/s
M2	**a**	u+5a=24, u+2a=12 → 3a=12, a=4; u=12−8=4
M3	**c**	Δv = 5×4 = 20; v = 6+20 = 26 m/s
M4	**c**	v_y = 0 + 10×3 = 30 m/s
M5	**b**	v = √(15²+30²) = √1125 ≈ 33.5 m/s
M6	**a**	u_x=40×0.866=34.6, u_y=40×0.5=20 m/s
M7	**b**	v_y = 20 − 10×1.5 = 5 m/s
M8	**b**	t = u_y/g = 20/10 = 2 s
M9	**b**	v_y = 20 − 10×3 = −10 → magnitude 10 m/s
M10	**b**	4t = 10 → t = 2.5 s
M11	**b**	v = 20 − 4×6 = −4 → 4 m/s down the incline
M12	**b**	Reaches 30 m/s at t=4s, then a=0 → stays 30 m/s at t=9s
M13	**b**	$\vec{a}$ rotates through the quarter turn, so $\vec{a}t$ (fixed direction) ≠ $\Delta\vec{v}$
M14	**a**	The 1-D form along the tangent needs constant tangential acceleration
H1	**a**	t_1t_2 = (2u sinθ/g)(2u cosθ/g) = 2u²sin2θ/g² = 2R/g
H2	**b**	u_y=25×0.8=20; half=10; 10=20−10t → t=1 s
H3	**a**	$\vec{a} = (\vec{v}-\vec{u})/t = (4\hat{i}+0\hat{j})/2 = 2\hat{i}$ m/s²
H4	**a**	v_x=3+2×5=13, v_y=4 (a_y=0) → $13\hat{i}+4\hat{j}$
H5	**b**	45° needs $\lvert v_y\rvert=v_x$=5 → 10t=5 → t=0.5 s
H6	**c**	5+2t=15 → t=5 s
H7	**a**	$v_{rel}=(u_P-u_Q)+a_{rel}t$ = −10+2×3 = −4 → P is 4 m/s slower (Q faster)
H8	**a**	Both true; R (gravity's horizontal component = 0) directly explains why components split cleanly
H9	**d**	A is false (equation invalid over an interval in UCM); R is a true fact about |a|
H10	**b**	The equation strictly needs $\vec{a}$ constant throughout — the final value alone isn't enough
H11	**c**	Both share a = −g, so relative velocity = $u_{thrown} - u_{dropped}$ = 20 m/s, constant for all t
H12	**a**	t ∝ sinθ → sin30°:sin60° = 0.5:0.866 = 1:√3
H13	**b**	$v_y(T) = u_y - g(2u_y/g) = -u_y$
H14	**a**	After 4s: (6,−5); then +2s of (0,1): (6, −5+2) = $6\hat{i}-3\hat{j}$